resources compared to qubits remains constant for the kinetic part, for the chromo-electric piece it is found to scale as $N_c(2N_c+17)/(3+3N_c)$, which increases as a function of N_c. Mapping fermion occupations to qudits, as we have presented in this work, inspired by quantum chemistry and nuclear many-body systems, are also expected to accelerate quantum simulations of quantum field theories in higher numbers of spatial dimensions. There, where both fermions and gauge bosons have to be considered, hybrid architectures might provide the most efficient mapping [54, 55]. This is the subject of future work.

ACKNOWLEDGMENTS

We would like to thank Crystal Senko and Sara Mouradian for discussions related to trapped-ion qudit systems which, in part, led to this work. We would also like to thank Roland Farrell for helpful discussions, and for all of our other colleagues and collaborators that provide the platform from which this work has emerged. Martin Savage would like to thank the Physics Department at Universität Bielefeld for kind hospitality during some of this work. This work was supported, in part, by Universität Bielefeld and ERC-885281-KILONOVA Advanced Grant (Caroline Robin), by U.S. Department of Energy, Office of Science, Office of Nuclear Physics, InQubator for Quantum Simulation (IQuS)[9] under DOE (NP) Award No. DE-SC0020970 via the program on Quantum Horizons: QIS Research and Innovation for Nuclear Science[10] (Martin Savage), and the Quantum Science Center (QSC),[11] a National Quantum Information Science Research Center of the U.S. Department of Energy (Marc Illa). This work was supported, in part, through the Department of Physics[12] and the College of Arts and Sciences[13] at the University of Washington. We have made extensive use of Wolfram `Mathematica` [95].

[9] https://iqus.uw.edu
[10] https://science.osti.gov/np/Research/Quantum-Information-Science
[11] https://qscience.org
[12] https://phys.washington.edu
[13] https://www.artsci.washington.edu

Appendix A: Gell-Mann Matrices

The matrix representation of the generators of SU(3) transformations, T^a, acting on the fundamental representation are related to the Gell-Mann matrices via $T^a = \frac{1}{2}\lambda^a$, such that $\mathrm{Tr}[T^a T^b] = \frac{1}{2}\delta^{ab}$. Using Gell-Mann's convention,

$$\lambda^1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \ , \ \lambda^2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \ , \ \lambda^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \ , \ \lambda^4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \ ,$$
$$\lambda^5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix} \ , \ \lambda^6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \ , \ \lambda^7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \ , \ \lambda^8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \ . \tag{A1}$$

Appendix B: Embedding Quarks and Anti-Quarks into Qu8its

The fully antisymmetric quark states can be built from the vacuum and fermionic creation operators. Starting with the vacuum state,

$$|1\rangle \equiv |\Omega\rangle \ , \tag{B1}$$

the one-quark states are,

$$\hat{c}_r^\dagger|1\rangle = \hat{c}_r^\dagger|\Omega\rangle = |q_r\rangle \equiv |2\rangle \ , \quad \hat{c}_g^\dagger|1\rangle = \hat{c}_g^\dagger|\Omega\rangle = |q_g\rangle \equiv |3\rangle \ , \quad \hat{c}_b^\dagger|1\rangle = \hat{c}_b^\dagger|\Omega\rangle = |q_b\rangle \equiv |4\rangle \ , \tag{B2}$$

the two-quark states are,

$$\begin{aligned}
&\hat{c}_r^\dagger|2\rangle = \hat{c}_r^\dagger|q_r\rangle = 0 \ , && \hat{c}_g^\dagger|2\rangle = \hat{c}_g^\dagger|q_r\rangle = \frac{1}{\sqrt{2}}|q_g q_r - q_r q_g\rangle \equiv -|7\rangle \ , \\
&\hat{c}_r^\dagger|3\rangle = \hat{c}_r^\dagger|q_g\rangle = \frac{1}{\sqrt{2}}|q_r q_g - q_g q_r\rangle \equiv |7\rangle \ , && \hat{c}_g^\dagger|3\rangle = \hat{c}_g^\dagger|q_g\rangle = 0 \ , \\
&\hat{c}_r^\dagger|4\rangle = \hat{c}_r^\dagger|q_b\rangle = \frac{1}{\sqrt{2}}|q_r q_b - q_b q_r\rangle \equiv -|6\rangle \ , && \hat{c}_g^\dagger|4\rangle = \hat{c}_g^\dagger|q_b\rangle = \frac{1}{\sqrt{2}}|q_g q_b - q_b q_g\rangle \equiv |5\rangle \ , \\
&\hat{c}_b^\dagger|2\rangle = \hat{c}_b^\dagger|q_r\rangle = \frac{1}{\sqrt{2}}|q_b q_r - q_r q_b\rangle \equiv |6\rangle \ , \\
&\hat{c}_b^\dagger|3\rangle = \hat{c}_b^\dagger|q_g\rangle = \frac{1}{\sqrt{2}}|q_b q_g - q_g q_b\rangle \equiv -|5\rangle \ , \\
&\hat{c}_b^\dagger|4\rangle = \hat{c}_b^\dagger|q_b\rangle = 0 \ ,
\end{aligned} \tag{B3}$$

and the three-quark state is,

$$\begin{aligned}
&\hat{c}_r^\dagger|5\rangle = \hat{c}_r^\dagger \frac{1}{\sqrt{2}}|q_g q_b - q_b q_g\rangle = \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{3}}|q_r q_g q_b - q_g q_r q_b + q_g q_b q_r\rangle - \frac{1}{\sqrt{3}}|q_r q_b q_g - q_b q_r q_g + q_b q_g q_r\rangle\right) \equiv |8\rangle \ , \\
&\hat{c}_r^\dagger|6\rangle = \hat{c}_r^\dagger \frac{1}{\sqrt{2}}|q_b q_r - q_r q_b\rangle = 0 \ , && \hat{c}_r^\dagger|7\rangle = \hat{c}_r^\dagger \frac{1}{\sqrt{2}}|q_r q_g - q_g q_r\rangle = 0 \ , \\
&\hat{c}_g^\dagger|5\rangle = \hat{c}_g^\dagger \frac{1}{\sqrt{2}}|q_g q_b - q_b q_g\rangle = 0 \ , && \hat{c}_b^\dagger|5\rangle = \hat{c}_b^\dagger \frac{1}{\sqrt{2}}|q_g q_b - q_b q_g\rangle = 0 \ , \\
&\hat{c}_g^\dagger|6\rangle = \hat{c}_g^\dagger \frac{1}{\sqrt{2}}|q_b q_r - q_r q_b\rangle \equiv |8\rangle \ , && \hat{c}_b^\dagger|6\rangle = \hat{c}_b^\dagger \frac{1}{\sqrt{2}}|q_b q_r - q_r q_b\rangle = 0 \ , \\
&\hat{c}_g^\dagger|7\rangle = \hat{c}_g^\dagger \frac{1}{\sqrt{2}}|q_r q_g - q_g q_r\rangle = 0 \ , && \hat{c}_b^\dagger|7\rangle = \hat{c}_b^\dagger \frac{1}{\sqrt{2}}|q_r q_g - q_g q_r\rangle \equiv |8\rangle \ .
\end{aligned} \tag{B4}$$

We note that with these definitions, it is equally profitable to denote the qu8it states as

$$\begin{aligned}\left\{|\text{qu8it}\rangle\right\} &= \left\{|\Omega\rangle\ ,\ |q_r\rangle\ ,\ |q_g\rangle\ ,\ |q_b\rangle\ ,\ |q_g q_b\rangle\ ,\ -|q_r q_b\rangle\ ,\ |q_r q_g\rangle\ ,\ |q_r q_g q_b\rangle\right\} \\ &= \left\{|1\rangle\ ,\ |2\rangle\ ,\ |3\rangle\ ,\ |4\rangle\ ,\ |5\rangle\ ,\ |6\rangle\ ,\ |7\rangle\ ,\ |8\rangle\right\} \\ &= \left\{|\mathbf{1}_0\rangle\ ,\ |\mathbf{3}_1,1\rangle\ ,\ |\mathbf{3}_1,2\rangle\ ,\ |\mathbf{3}_1,3\rangle\ ,\ |\overline{\mathbf{3}}_2,1\rangle\ ,\ |\overline{\mathbf{3}}_2,2\rangle\ ,\ |\overline{\mathbf{3}}_2,3\rangle\ ,\ |\mathbf{1}_3\rangle\right\}\ ,\end{aligned} \tag{B5}$$

where the subindex in the irrep labels the number of quarks in the state. As discussed in the main text, an analogous mapping for the anti-quarks is,

$$\begin{aligned}\left\{|\overline{\text{qu8it}}\rangle\right\} &= \left\{|\phi\rangle\ ,\ |\overline{q}_r\rangle\ ,\ |\overline{q}_g\rangle\ ,\ |\overline{q}_b\rangle\ ,\ |\overline{q}_g\overline{q}_b\rangle\ ,\ -|\overline{q}_r\overline{q}_b\rangle\ ,\ |\overline{q}_r\overline{q}_g\rangle\ ,\ |\overline{q}_r\overline{q}_g\overline{q}_b\rangle\right\} \\ &= \left\{|\bar{1}\rangle\ ,\ |\bar{2}\rangle\ ,\ |\bar{3}\rangle\ ,\ |\bar{4}\rangle\ ,\ |\bar{5}\rangle\ ,\ |\bar{6}\rangle\ ,\ |\bar{7}\rangle\ ,\ |\bar{8}\rangle\right\} \\ &= \left\{|\mathbf{1}_0\rangle\ ,\ |\overline{\mathbf{3}}_1,1\rangle\ ,\ |\overline{\mathbf{3}}_1,2\rangle\ ,\ |\overline{\mathbf{3}}_1,3\rangle\ ,\ |\mathbf{3}_2,1\rangle\ ,\ |\mathbf{3}_2,2\rangle\ ,\ |\mathbf{3}_2,3\rangle\ ,\ |\mathbf{1}_3\rangle\right\}\ .\end{aligned} \tag{B6}$$

Appendix C: Givens Rotations for SU(8)

Givens rotations are a straightforward way to access SU(8) transformations, and are particularly convenient for quantum operations that are sequential applications of two-level transformations, as are induced, for example, by application of lasers to trapped ions. The notation that we will use parallels and extends the notation used for Pauli operators, σ_x, σ_y and $\sigma_z \to \mathcal{X}_{ij}$, $\mathcal{Y}_{ij}$ and $\mathcal{Z}_i$. For SU(8) transformations, there are 28 $\mathcal{X}_{ij}$s, 28 $\mathcal{Y}_{ij}$s, and 7 $\mathcal{Z}_i$s.

For the 7 diagonal generators $\mathcal{Z}_i$s, one basis is a straightforward extension of Gell-Mann's SU(3):

$$\begin{aligned}\mathcal{Z}_1 &= \text{diag}\,(1,-1,0,0,0,0,0,0)\ ,\quad \mathcal{Z}_2 = \frac{1}{\sqrt{3}}\text{diag}\,(1,1,-2,0,0,0,0,0)\ , \\ \mathcal{Z}_3 &= \frac{1}{\sqrt{6}}\text{diag}\,(1,1,1,-3,0,0,0,0)\ ,\quad \mathcal{Z}_4 = \frac{1}{\sqrt{10}}\text{diag}\,(1,1,1,1,-4,0,0,0)\ , \\ \mathcal{Z}_5 &= \frac{1}{\sqrt{15}}\text{diag}\,(1,1,1,1,1,-5,0,0)\ ,\quad \mathcal{Z}_6 = \frac{1}{\sqrt{21}}\text{diag}\,(1,1,1,1,1,1,-6,0)\ , \\ \mathcal{Z}_7 &= \frac{1}{\sqrt{28}}\text{diag}\,(1,1,1,1,1,1,1,-7)\ ,\quad \text{Tr}\,[\mathcal{Z}_i\mathcal{Z}_j] = 2\delta^{ij}\ .\end{aligned} \tag{C1}$$

However, an alternate choice that makes better connection to sequency analysis, and which we have chosen to use in our work, is the Hadamard-Walsh basis, which includes the identity operator,

$$\begin{aligned}w_1 &= \frac{1}{\sqrt{8}}\text{diag}\,(1,1,1,1,1,1,1,1)\ ,\quad w_2 = \frac{1}{\sqrt{8}}\text{diag}\,(1,1,1,1,-1,-1,-1,-1)\ , \\ w_3 &= \frac{1}{\sqrt{8}}\text{diag}\,(1,1,-1,-1,-1,-1,1,1)\ ,\quad w_4 = \frac{1}{\sqrt{8}}\text{diag}\,(1,1,-1,-1,1,1,-1,-1)\ , \\ w_5 &= \frac{1}{\sqrt{8}}\text{diag}\,(1,-1,-1,1,1,-1,-1,1)\ ,\quad w_6 = \frac{1}{\sqrt{8}}\text{diag}\,(1,-1,-1,1,-1,1,1,-1)\ , \\ w_7 &= \frac{1}{\sqrt{8}}\text{diag}\,(1,-1,1,-1,-1,1,-1,1)\ ,\quad w_8 = \frac{1}{\sqrt{8}}\text{diag}\,(1,-1,1,-1,1,-1,1,-1)\ ,\end{aligned} \tag{C2}$$

which are normalized such that $\text{Tr}\,[w_i w_j] = \delta^{ij}$. The ordering of the w_i is the same as those produced in `Mathematica` from `HadamardMatrix[8]`.

The matrix representations of the (symmetric) σ^x-type generators, $\mathcal{X}_{ij}$, have all zero entries except for the elements defined by ij, such that $\mathcal{X}_{ij} = \mathcal{X}_{ji} = 1$. Similarly, the matrix representations of the (anti-symmetric) σ^y-type